MIL-M-8917A(ASG)
2 MAY 1969

Superseding
MIL-M-8917(ASG)
28 March 1962

MILITARY SPECIFICATION

MAGNESIUM ALLOY SHEET AND PLATE (HM21A)

This specification has been approved by the Department of the Air Force and by the Naval Air Systems Command.

1. SCOPE

* 1.1 This specification covers HM21A magnesium alloy sheet and plate.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-M-3171	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on

STANDARDS

Federal

FED. TEST METHOD STD. NO. 151	Metals; Test Methods
FED. STD. NO. 184	Identification Marking of Aluminum, Magnesium and Titanium
FED. STD. NO. 245	Tolerances for Aluminum Alloy and Magnesium Alloy Wrought Products

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-649	Aluminum and Magnesium Products, Preparation for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9535

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* 2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

American Society for Testing and Materials

ASTM E8 - 66 Tension Testing of Metallic Materials
ASTM E21 - 66T Short Time Elevated Temperature Tension Tests of Materials

(Copies of ASTM Standards may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Atomic Energy Commission

Code of Federal Regulations - Title 10, Part 20 - Standards for Protection Against Radiation

(Application for copies should be addressed to the Division of Licensing and Regulation, U.S. Atomic Energy Commission, Washington, D.C. 20545.)

3. REQUIREMENTS

3.1 Material. - The material shall be such as to produce sheet and plate conforming to this specification.

* 3.2 Chemical composition. - When tested in accordance with 4.4.2, the chemical composition shall be as specified in table I.

TABLE I. Chemical composition

Element	Percent by weight	
	Min.	Max.
Thorium	1.5	2.5
Manganese	0.45	1.1
Other impurities, each	-	1/ 0.10
Other impurities, total	-	1/ 0.30
Magnesium	Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated or suspected in amounts greater than specified limits, further analysis shall be made to determine that these elements are not present in excess of the specified limits.

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3.2.1 The contractor shall furnish an analysis of each melt represented by the lot showing the percentage of each of the elements specified in table I.

3.2.3 Chemical analysis by the contractor of the individual melts may be waived at the discretion of the procuring activity, provided that the producer's method of composition control is acceptable or that all material in the lot can be identified as being from melts previously analyzed and found to be in conformance with the chemical composition requirements.

* 3.3 Mechanical properties. - When tested as specified in 4.5.2, the mechanical properties of the sheet and plate, measured in the longitudinal (the direction of rolling) or long transverse direction shall conform to table II.

TABLE II. Mechanical properties of HM21A-T8

Alloy and temper	Specified thickness (inch)	Tensile strength, minimum (psi)	Yield strength at 0.2 percent offset, minimum (psi)	Elongation, percent in 2 inches, or 4D minimum
HM21A-T8	0.016-0.250 0.251-0.500 0.501-3.000	33,000 32,000 30,000	18,000 21,000 21,000	6 6 6
HM21A-T81	0.125-0.312	34,000	25,000	4

3.3.1 Minimum mechanical properties for thicknesses not shown in table II shall be agreed upon between the supplier and the procuring activity (see 6.2).

* 3.3.2 Tensile properties at 600° F. - When tested as specified in 4.5.2.2, material from each lot shall be capable of meeting the minimum mechanical properties of table III at 600° F.

TABLE III. Minimum mechanical properties at 600° F (HM21A-T8)

Alloy and temper	Specified thickness (inch)	Tensile strength, minimum (psi)	Elongation, percent in 2 inches, or 4D minimum
HM21A-T8	0.016-0.250	11,000	8
HM21A-T81	0.125-0.312	15,000	-

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3.4. Dimensional tolerances. - Magnesium alloy sheet and plate shall not vary from the specified dimensions or from true straightness of edge by an amount greater than the tolerances specified for flatness, squareness, thickness, width, length, and lateral bow in Federal Standard No. 245 for plate, flat sheet, or coiled sheet, as applicable.

3.4.1 Tolerances for sizes not covered by Federal Standard No. 245 shall be as specified in the contract or order (see 6.2).

3.5 Marking for identification. - Unless otherwise specified in the contract or order (see 6.2), marking shall be in accordance with Federal Standard No. 184. In addition to the commercial alloy designation, the number of this specification shall be applied to each sheet and plate.

3.6 Precautionary information. - The processing, handling, and shipment of products covered by this specification shall be as specified in the Code of Federal Regulations, Title 10, Part 20, titled "Standards for Protection Against Radiation" (see 6.2).

3.7 Protective treatment. - Sheet and plate up through 0.500 inch thick shall be acid pickled and then either oiled or chrome pickled, as specified (see 6.2). Plate over 0.500 inch thick shall be either oiled or chrome pickled as specified (see 6.2) as a final process before shipment. Unless otherwise specified by the procuring activity (see 6.2), the sheet or plate shall be given a chrome pickle treatment in accordance with type I of MIL-M-3171.

3.8 Workmanship. - Sheet and plate shall be uniform in quality and condition, and shall be clean, sound, smooth, commercially flat, and free from severe buckles and other injurious defects. Discoloration resulting from annealing operations, and superficial surface abrasions which are not of sufficient depth to affect adversely the static or dynamic mechanical properties or formability of the material, will not be cause for rejection.

3.8.1 The material shall be given such uniform treatment as the manufacturer desires to produce material that will conform to this specification, applied in a manner that will produce the utmost uniformity. The treatment shall be performed on the whole product, never on a part of the product only.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Lot. - A lot shall consist of material of the same condition and thickness, manufactured during the same production period.

4.3 Examination. - Each sheet and plate shall be carefully examined to determine conformance with this specification with respect to workmanship and marking. Sufficient spot checks shall be made to insure conformance with the tolerances specified in 3.4. On approval, by the procuring activity, a system of statistical quality control may be used for dimensional and workmanship examination.

4.4 Chemical analysis. -

4.4.1 Sampling. - A sample for chemical analysis shall be taken from each 4,000 pounds or less of material comprising the lot. When the sample for analysis is taken from the fabricated product, it shall consist of equal quantities of chips taken from at least four representative pieces of the full thickness of the material.

4.4.2 Method of analysis. - The sample selected shall be analyzed by the wet chemical method in accordance with Method 111 of Federal Test Method Standard No. 151, or the spectrographic method in accordance with Method 112 of Federal Test Method Standard No. 151 to determine conformance with 3.2. A single analysis of a composite sample may be made. In case of dispute, analysis by the wet method (Method 111) shall be the basis for acceptance.

4.5 Mechanical properties (tension test). -

4.5.1 Sampling. - At least one test sample shall be selected to represent each 1,000 pounds or less of sheet, or 2,000 pounds or less of plate comprising a lot, except that in no case shall less than two tests represent a lot. The test samples that represent a lot shall not be selected from the same sheet or plate.

* 4.5.1.1 Location of test specimens. - Test specimens of the full thickness of the material shall be used up to 0.500 inch in thickness; the specimen shall be taken midway between the two plate surfaces for plate in thicknesses of 0.500 inch up through 1.500 inch, and midway between the center and the surface (quarter point) of plate over 1.500 inch in thickness.

* 4.5.1.2 Specimens for room temperature tension test. - For material less than 0.500 inch in thickness, the tension test specimen shall be the rectangular specimen in ASTM E8. For material 0.500 inch or more in thickness, the tension test specimen shall be the round tension test specimen as illustrated in ASTM E8.

* 4.5.1.3 Specimens for elevated temperature tension test. - Test specimens shall be the rectangular specimen of ASTM E8 and shall be cut parallel to the direction of rolling. Elongation requirements apply only to sheet 3/4 inch and over in width.

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4.5.2 Tension tests.-

4.5.2.1 Room temperature. - The test shall be performed in accordance with ASTM E8.

* 4.5.2.2 Elevated temperature. - The test shall be performed in accordance with ASTM E21. Unless otherwise specified (see 6.2), test temperature shall be 600° F.

4.5.3 Yield strength. - Yield strength shall be determined by the offset method in accordance with ASTM E8.

4.6 Rejection and retests. -

4.6.1 Rejection. - If any individual test specimen fails to meet the requirements of this specification, the lot represented by the specimen shall be rejected, except when a statistical sampling plan is used.

4.6.2 Retests. - If a lot of material is rejected in accordance with 4.6.1, retests of that lot are permitted in accordance with the provisions of Federal Test Method Standard No. 151 by (a) taking twice the original number of test specimens from an area in the original sample adjacent to the area represented by the failure, or (b) taking the original number of test specimens from an area in each of two other sheets or plates in the lot. Should any of these specimens fail to meet the requirements of this specification, the entire lot shall be rejected.

4.7 Packaging, packing, and marking. - Preparation for delivery shall be examined for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation. - Unless otherwise specified, all material shall be separated by condition, size, and thickness. Individual sheets and plates which have been given a chrome pickle surface treatment shall be separated by a noncorrosive, nonhygroscopic paper to protect the finish.

5.2 Packing and packaging (see 3.6). - Unless otherwise specified by the procuring activity (see 6.2), material shall be packed in accordance with MIL-STD-649. Unless otherwise specified by the procuring activity (see 6.2), level C packaging and packing shall be used.

5.3 Marking for shipment. - In addition to any special marking required by the contract or order (see 6.2), marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. - HM21A sheet and plate is intended for use in components where weldability and good strength-to-weight ratio up to 600° F are required.

* 6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and thickness (see 3.3). Minimum mechanical properties for material of thickness not specified herein (see 3.3.1).
(c) Tolerances for size of material not covered by this specification (see 3.4.1).
(d) Any special marking required (see 3.5 and 5.3).
(e) Precautionary information (and Atomic Energy Commission license number) (see 3.6).
(f) Protective treatment (see 3.7)
(g) Elevated temperature for tension test (see 4.5.2.2).
(h) Preparation for delivery data (see section 5).

* 6.3 Marginal indicia. - The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the past previous issue.

Custodians:
Navy - AS
Air Force - 11

Preparing activity:
Air Force - 11

Project No. 9535-0194

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget
Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-M-8917A(ASG) MAGNESIUM ALLOY SHEET AND PLATE (HM21A)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

( ) Direct Government Contract ( ) Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?

   ( ) YES ( ) NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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